

Beyond the GPU

Driving factors, architectures, and economics of AI hardware acceleration — a systems lecture

In-depth lecture • ~2 hours

Seven movements

1

Why offload

The physical driving factors.

2

Workload pressures

The measured bounds.

3

The framework

Contracts + necessity test.

4

The families

Eight contract changes.

5

Landscape

Grouped, not cataloged.

6

Substrate

Software, packaging, cost.

7

Futures

Graded by evidence.

★

Conclusions

What holds everywhere.

The hardware vocabulary, in software terms



GPU

Thousands of parallel lanes; fast on regular, batched, streamed work.



Matrix / tensor engine

A whole matrix tile per instruction; control cost amortized over many multiply-adds.



Systolic array

A grid where operands flow to neighbors — a value fetched once is reused many times.



HBM & scratchpad

Stacked DRAM for bandwidth; software-managed SRAM for reuse under compiler control.



NPU / dataflow / PIM

Blocks that change a contract: wake-scope energy, static schedule, or compute in memory.



TOPS vs contract

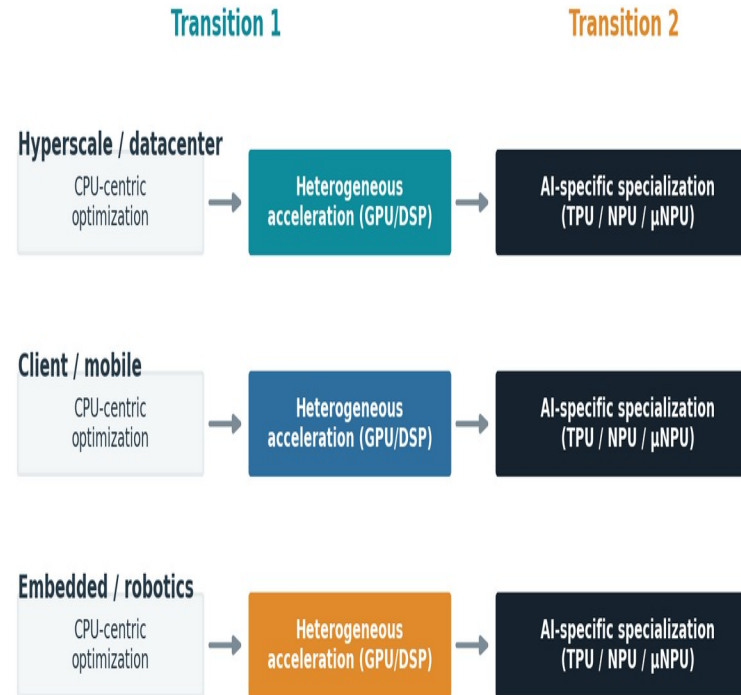
TOPS is a peak rating (marketing). A contract is what hardware promises software.

01

Why work leaves the GPU

The physical pressures behind offload

Two transitions, across the whole continuum



⇒ Software succeeds until it hits a limit; each transition adds specialization rather than replacing what came before.

Why general-purpose scaling stalled



The memory wall

Compute outran memory; the question became which data is resident and reused, not raw latency.



The power wall

Frequency scaling ended; more transistors help only when software exposes parallelism.



Dark silicon

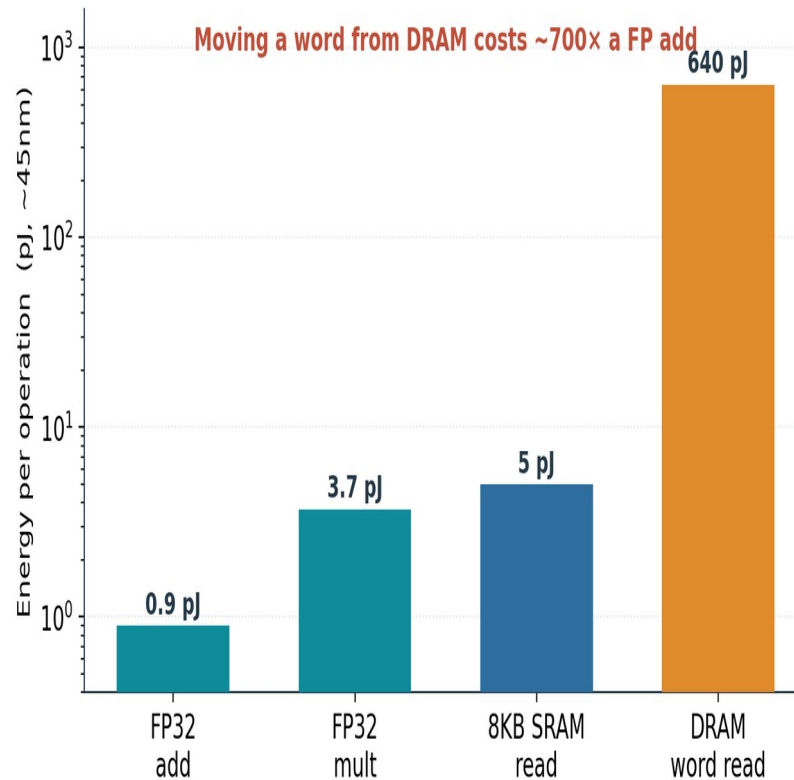
Not all transistors can switch at once in a fixed power budget — specialization does more per joule.



Data movement

Energy is dominated by moving operands, not computing on them — the recurring lever.

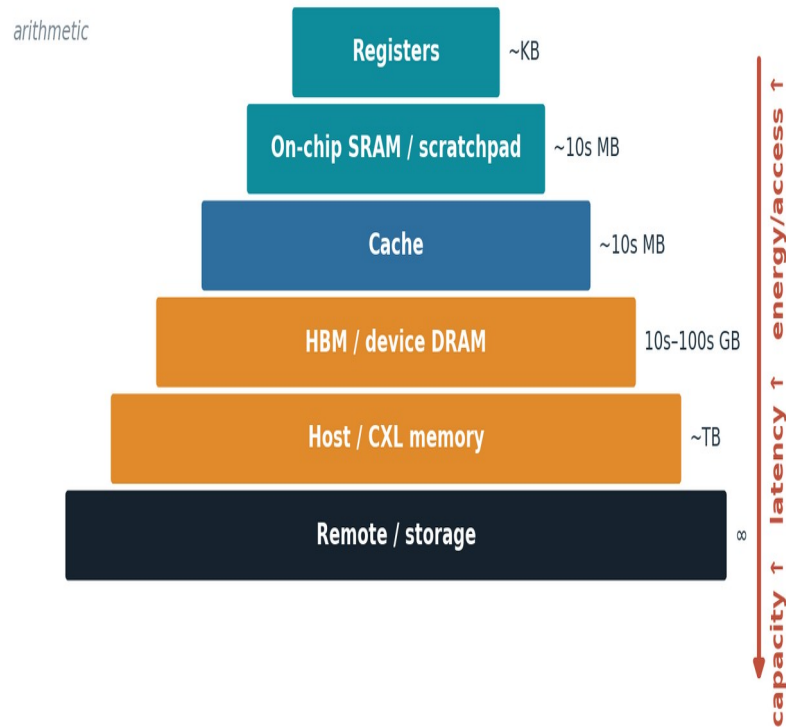
Data movement dominates energy



- Compute is nearly free.
- Moving operands is not.
- Locality sets efficiency, not FLOPS.

⇒ Every family that follows is a different way to move less data, or move it a shorter distance.

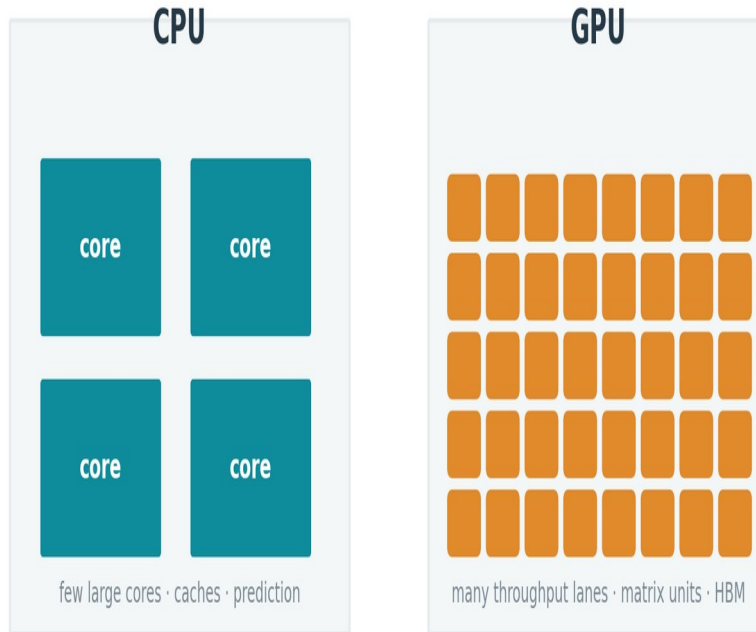
Memory is a distance hierarchy



- Farther = more capacity.
- Also more latency and energy.
- Caches help only with reuse.

⇒ 'Memory-bound' is many distinct problems — each a lever a specialized design can pull.

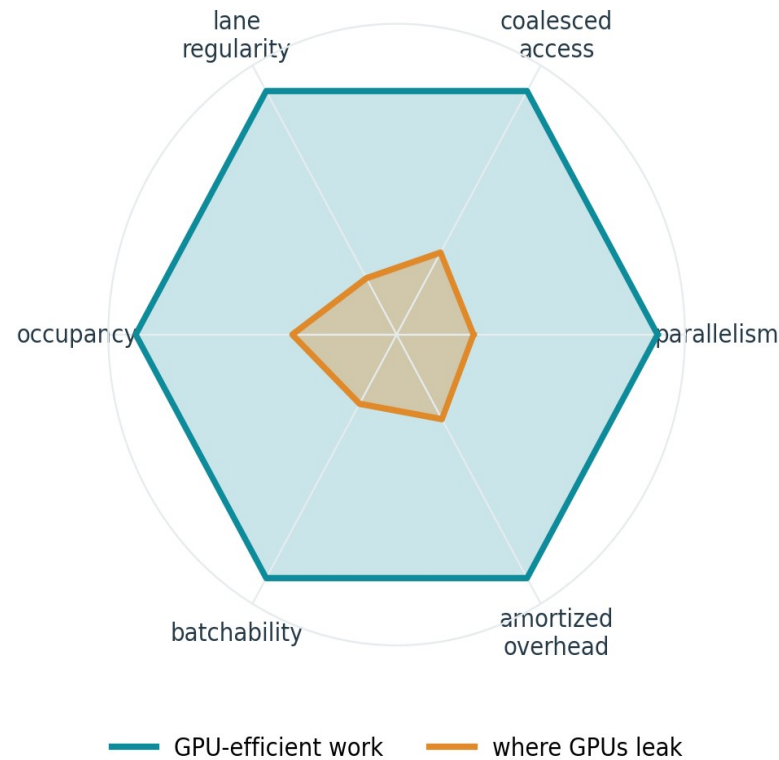
CPU and GPU spend silicon differently



- CPU: latency across diverse code.
- GPU: throughput across regular work.
- Neither wins every phase.

⇒ Specialization narrows the job in exchange for more useful work per area and per joule.

The GPU efficiency envelope is real and bounded



⇒ GPUs are unbeatable inside the envelope and leak outside it — that boundary is where specialists live.

02

The workload pressures

The measured bounds that reshape silicon

Signatures, not labels

Workload signatures, not labels

	arith. intensity	locality	parallelism	regularity	batchable	deadline
Training matmul	l	l	l	l	l	-
Transformer decode	-	lo	lo	l	lo	lo
Embedding lookup	-	-	l	-	hi	-
Graph aggregation	-	-	hi	-	lo	lo
Robotic planning	lo	lo	lo	lo	-	l

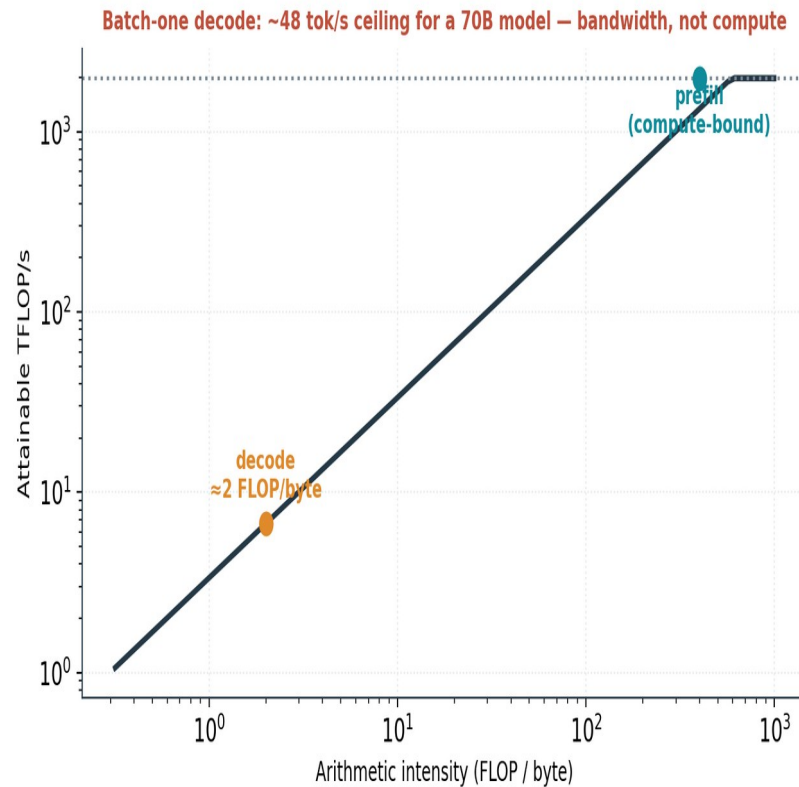
⇒ Characterize each phase by its behavior — the same model is compute-, bandwidth-, and capacity-bound in one request.

Prefill and decode want different machines



⇒ One pooled device compromises; phase-specialized silicon is now reaching hyperscale roadmaps.

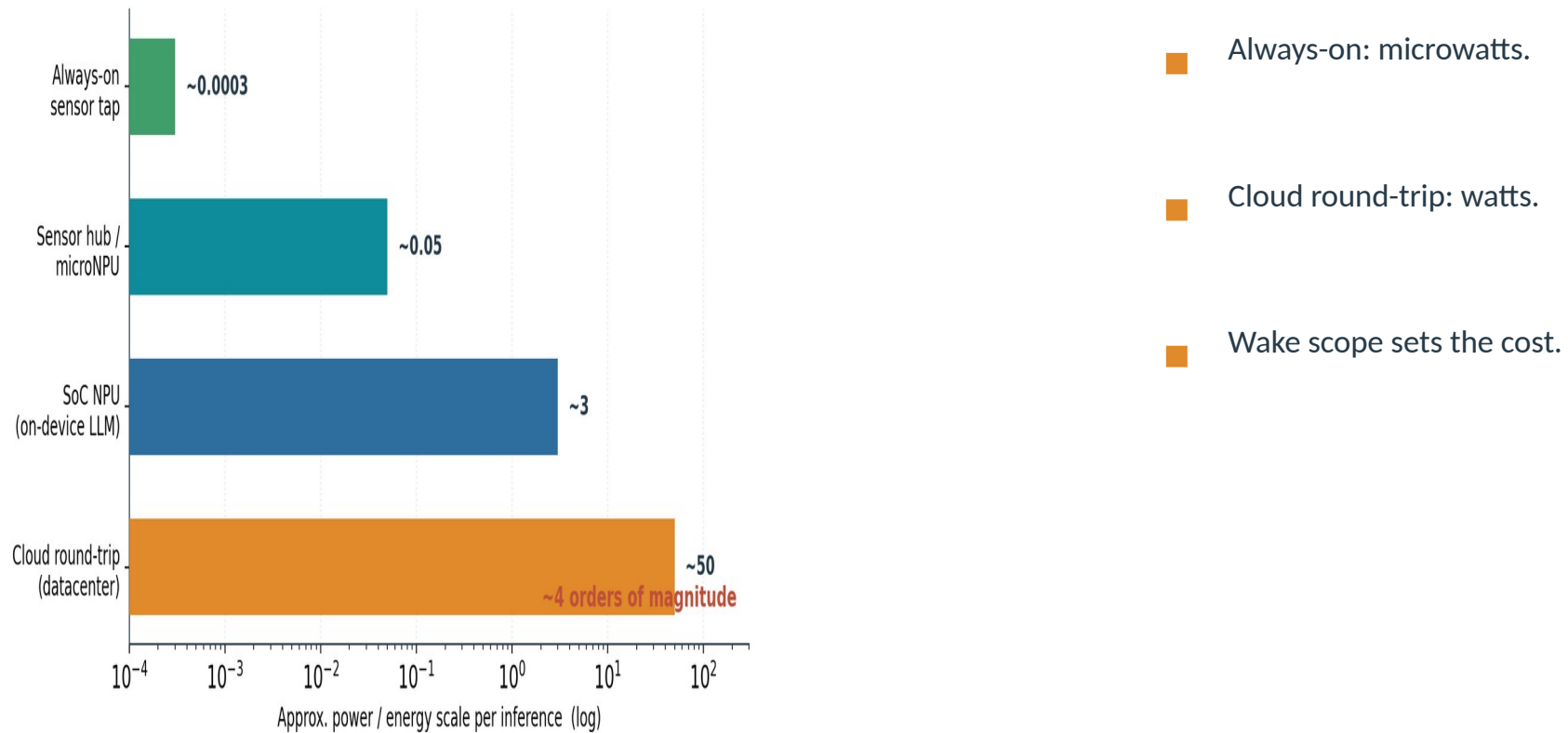
Batch-one decode is a bandwidth wall



- 70B model, 8-bit, one sequence.
- ≈48 tokens/s ceiling on an H100.
- ~0.3% of matrix throughput used.

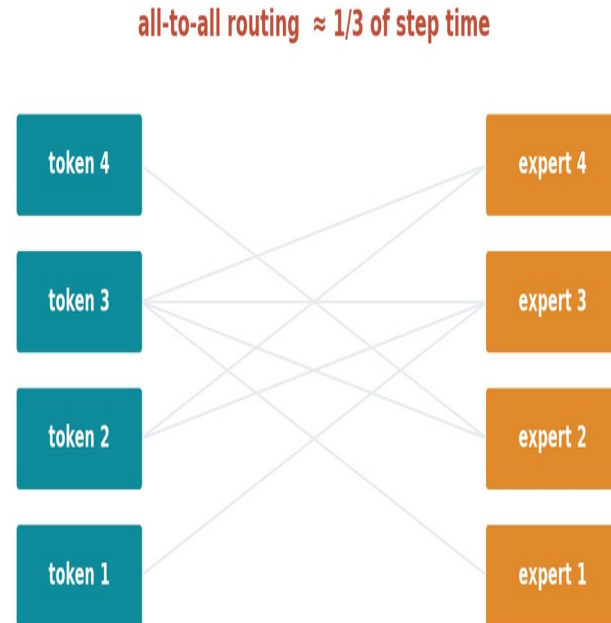
⇒ Only batching or a changed memory contract moves this — the pressure behind inference-first silicon.

Energy spans four orders of magnitude



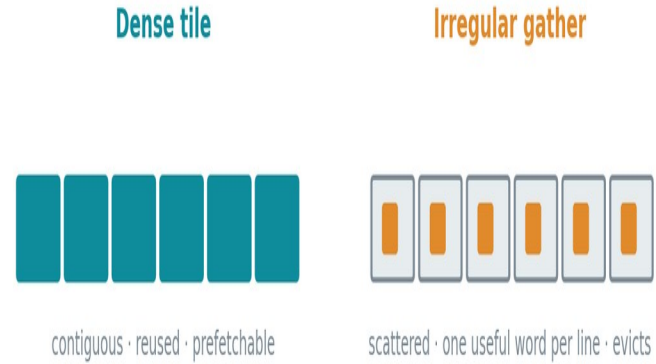
⇒ Always-on features are impossible on a GPU contract — a structural bound, not an efficiency gap.

MoE turns compute into communication



⇒ All-to-all routing can be $\sim 1/3$ of a training step — a bound in the fabric, not the arithmetic.

Irregular access defeats the cache



⇒ Embeddings, gather, and graph traversal are the bound no family yet owns cleanly.

Software succeeds — until it meets physics

3×

IO-aware attention throughput
(FlashAttention)

2–4×

KV paging / continuous batching
(vLLM)

~34%

of a training step in MoE all-to-all

<300 μ W

always-on keyword spotting
budget

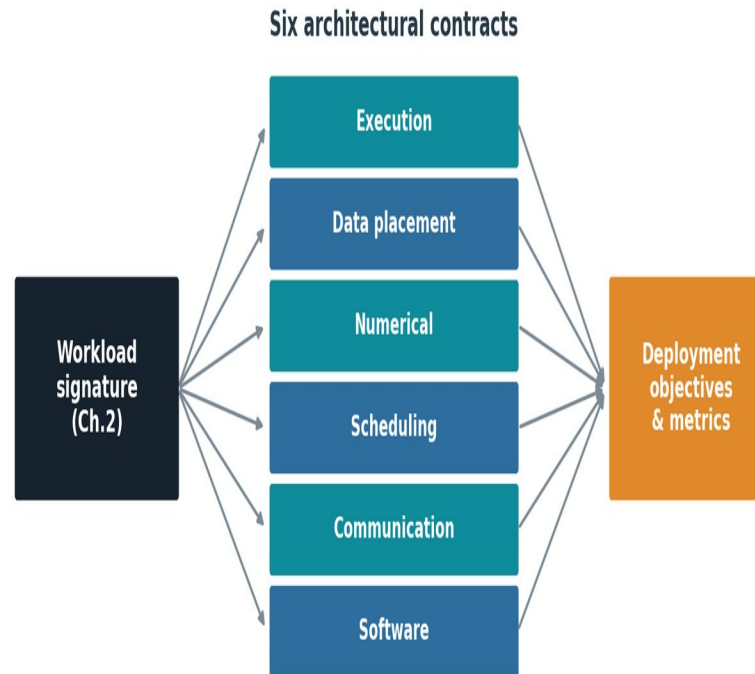
⇒ Each gain is real and each exposes a residual bound software cannot move — that residue is the case for hardware.

03

The decision framework

Contracts, and a test for when to specialize

Hardware defines six contracts



⇒ A signature stresses some contracts, not others — that mismatch is where a new architecture earns its place.

Four conditions — all must hold

1

Residual bound

A limit remains after software has done its best.

2

Material consequence

The bound actually hurts under the deployment's objective.

3

Effective lever

Hardware changes the bound rather than relocating it.

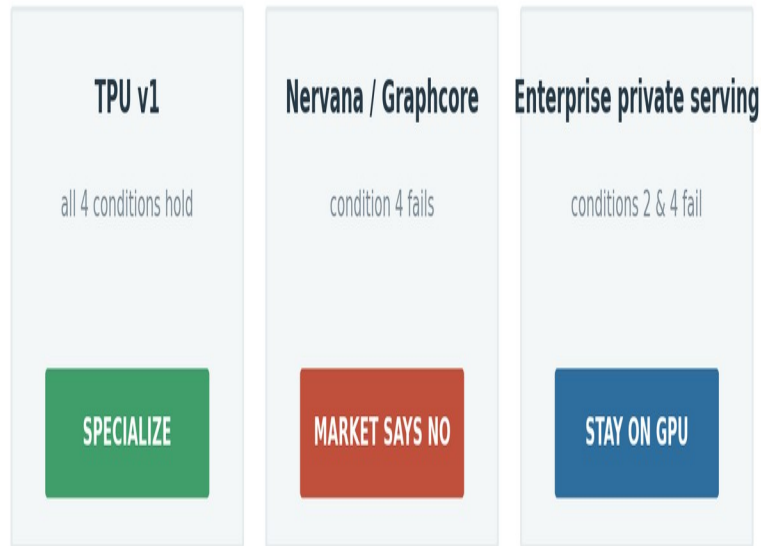
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Lifetime economics

Recurring benefit repays fixed cost, software, and lost flexibility.

⇒ Conditions 1-3 are the architect's analysis; condition 4 is the business case — and it decides most outcomes.

Same architecture, opposite verdicts



⇒ TPU v1 specialized; Nervana/Graphcore were right on architecture but failed condition 4; enterprise stays on GPU.

What the GPU absorbs — and what it cannot

Figure 5.5 — Partition of residual bounds by absorbability
Residual bounds from Chapter 2, partitioned by whether GPU evolution can absorb them without abandoning the platform contract (host-driven submission, SIMT programmability, shared software capital)

ABSORBED INTO THE GPU PLATFORM (evidence class A)	CONTESTED MIDDLE (decided by §2.14 condition 4)	STRUCTURALLY OUTSIDE THE THROUGHPUT CONTRACT
Matrix density → tensor cores	Low-batch decode bandwidth bound	Wake-scope / always-on energy (μW vs W domains)
Precision → FP16/BF16/FP8 engines	MoE routing / all-to-all ($\sim 1/3$ step time)	Provable worst-case timing and certification
Structured sparsity → 2:4 paths	Near-memory embedding lookup	Near-sensor locality
Movement overhead → async copy / TMA	Phase-disaggregated serving	Embedded per-unit cost floors
Launch overhead → graph capture	GPU trajectory + software capital vs. contract-changing designs	Secure basis of the beyond-GPU case
Scale-up comms → NVLink domains		
Isolation → device partitioning		

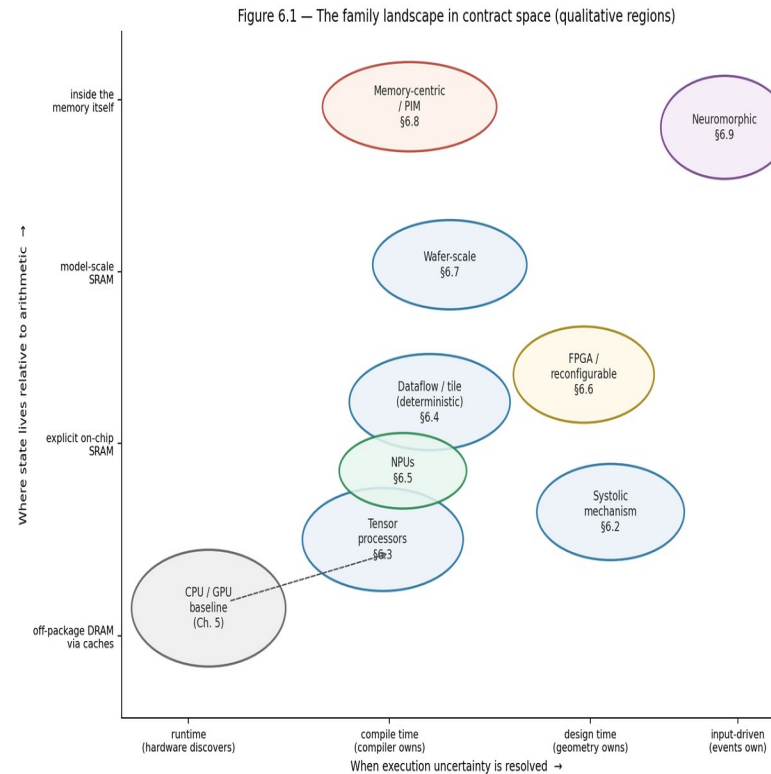
⇒ Absorbable bounds join the platform; structural bounds (wake scope, determinism, near-sensor, cost floors) do not.

04

The accelerator families

Eight contract changes, two organizing axes

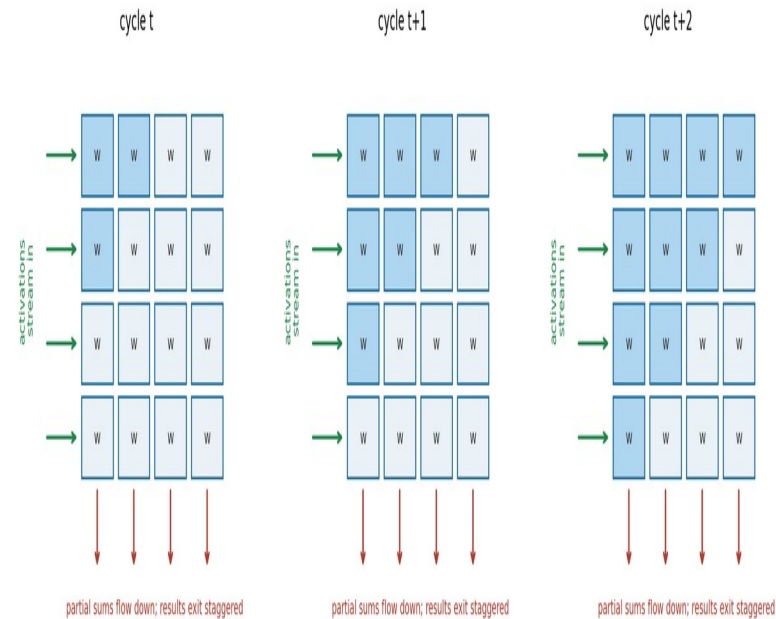
When uncertainty resolves × where state lives



⇒ Every family is a displacement from the CPU/GPU baseline on these two axes — no point dominates.

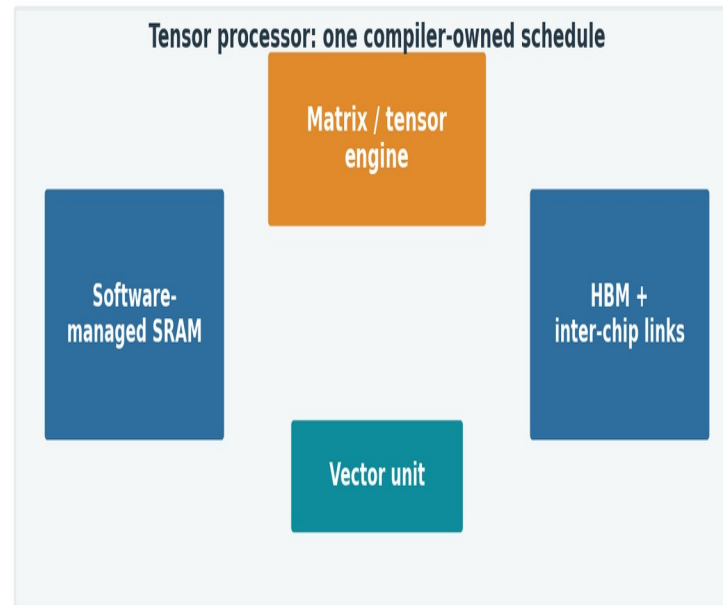
Systolic arrays: reuse from geometry

Figure 6.2 — Weight-stationary systolic execution: weights resident (w), activation wavefront advances one hop per cycle; every fetched value is reused across the array



⇒ A value fetched once is reused across the array — memory traffic only at the edges. Absorbed by everyone, including the GPU.

Tensor processors: one compiler-owned schedule



⇒ **Matrix engine + software-managed SRAM. Wins dense portfolios at owner scale (TPU, Trainium, MTIA).**

Deterministic dataflow / tile processors

GPU: uncertainty discovered at run time

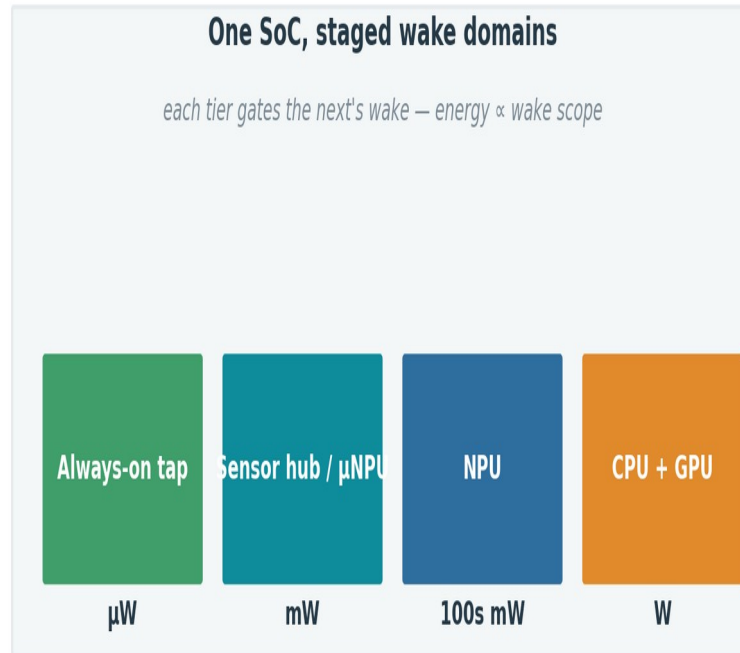


Dataflow / tile: schedule fixed at compile time



⇒ Schedule fixed at compile time → zero variance. Bids for latency-bounded decode and reproducible execution (Groq).

NPU is the client's standard low-power AI block



- Universal in client SoCs (≥ 40 TOPS floor).
- Power-domain-first; sustained/ambient work.
- Heavy generative inference still runs on the GPU.
- Coverage, not TOPS, decides value.

⇒ **Structural claim:** wake-scope energy + cost floors sit outside the GPU contract, so the edge is where specialization is safest — not that NPUs win the client outright.

FPGAs: flexibility after manufacturing

Reconfigurable fabric: two compilation speeds

Hardware compile

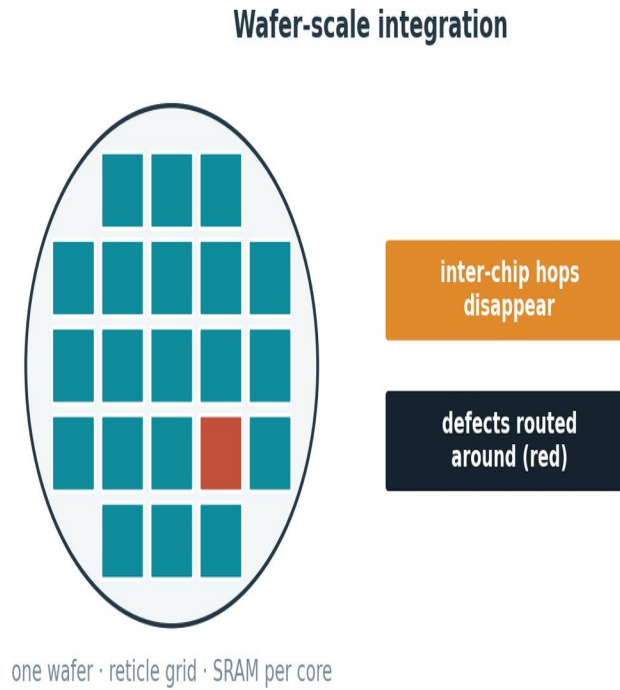
hours · max efficiency
hardware skills

Overlay / soft NPU

seconds · lower efficiency
software skills

⇒ Two compilation speeds. Wins where post-manufacture structure is the requirement; elsewhere a staging tier to ASIC.

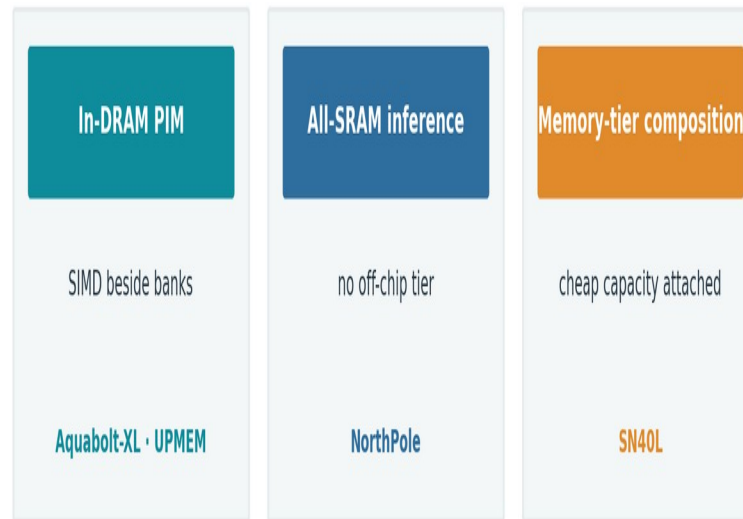
Wafer-scale: state at SRAM distance



⇒ Inter-chip communication becomes on-wafer routing; model-scale state stays close. Survives on anchor tenants.

Memory-centric / processing-in-memory

Compute moves to the data – three flavors



⇒ Strongest condition-3 case in the survey — gated only by the lack of a portable programming model.

Neuromorphic: energy proportional to activity



Mechanism

Event-driven spikes; synaptic state beside the neurons; nothing switches when nothing changes.



Evidence

Research-class (TrueNorth, Loihi). Validates the energy model, not yet a mainstream workload.

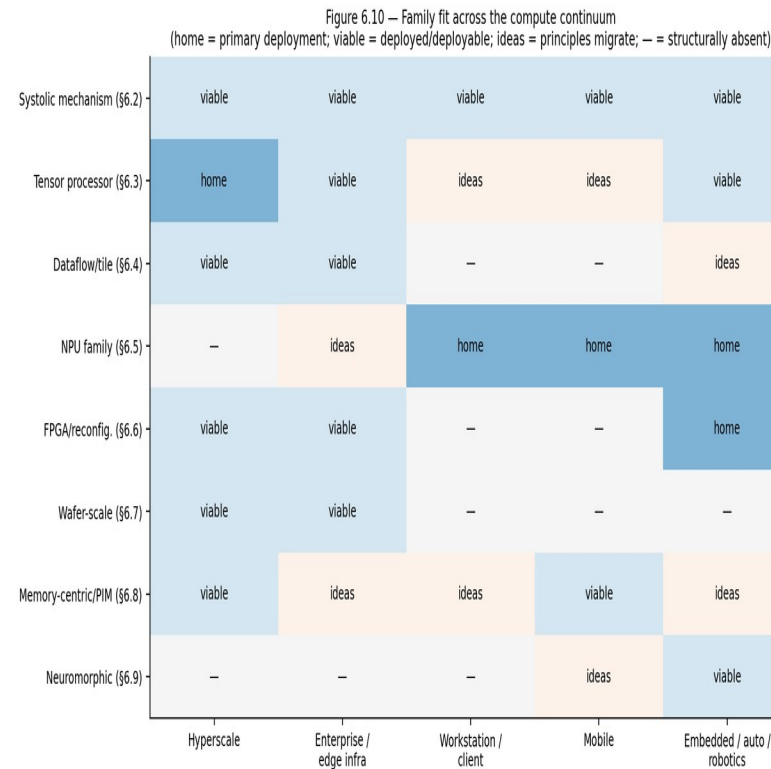


Beachhead

Extreme edge: always-on sensory, reflex loops, event cameras — sparse-in-time signals.

⇒ The theoretical endpoint of wake-scope logic — a direction of development, not yet a destination.

Same families, different binding constraints



⇒ Ideas migrate both ways; implementations diverge because each deployment prices the stressors differently.

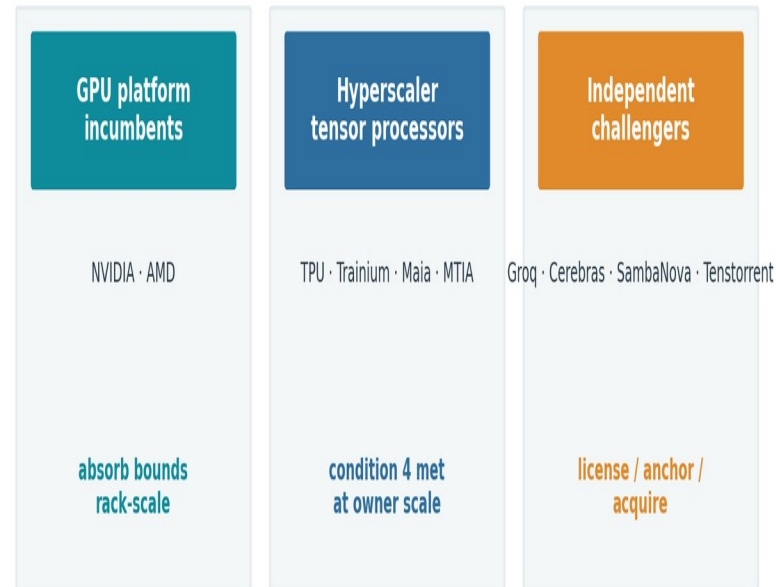
05

The landscape, grouped

Read through the framework — not a product catalog

Three groups, one deciding variable

Grouped by how condition 4 resolves — not a product catalog



⇒ None of the challengers failed on architecture — every trajectory was decided by condition 4.

The TPU verdict, replicated four times



Google TPU

Seventh generation; inference-emphasized; pods of thousands.



AWS Trainium

Generally available; proprietary switch fabric; 'token economics.'



Microsoft Maia

Inference-first; large on-chip SRAM; standard Ethernet.

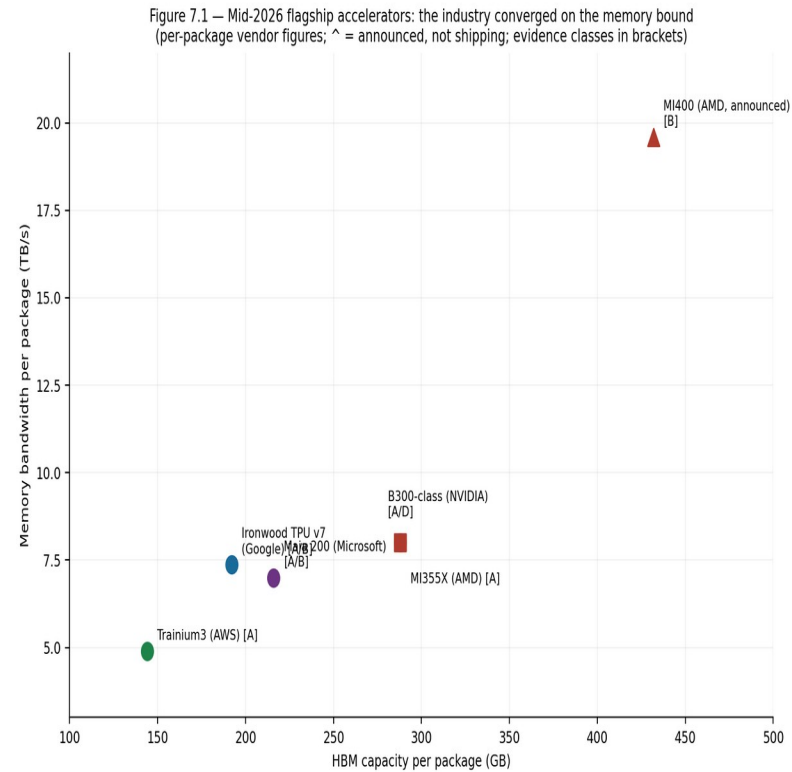


Meta MTIA

Recommendation-first; peer-reviewed first generation.

⇒ Every major cloud fields internal tensor processors at generation depth — the second transition as industrial fact.

The competition is a memory race



⇒ Flagships cluster on capacity and bandwidth, not FLOPS — the absorption thesis operating in public.

The contested middle consolidates



License in

A deterministic-dataflow architecture licensed into the incumbent platform.



Anchor tenancy

Wafer-scale survives on a few enormous compute agreements, then IPO.



Acquisition

Programmable tile processors absorbed by a larger incumbent.

⇒ **Condition 4 — software capital, volume, distribution — decided each, not architectural merit.**

The continuum's other half



AI PC

NPU + GPU under shared memory; bandwidth and coverage, not TOPS, decide.



Mobile SoC

Three wake tiers; on-device foundation models; thermal equilibrium separates the field.



Embedded / μ NPU

IP-licensed; cost floor and decade lifecycle dominate.



Robotics

GPU-class perception above a deterministic control floor — flexibility wins when algorithms churn.



Automotive

Determinism and lifecycle become certification objects; supply sovereignty joins economics.



Shared taxonomy

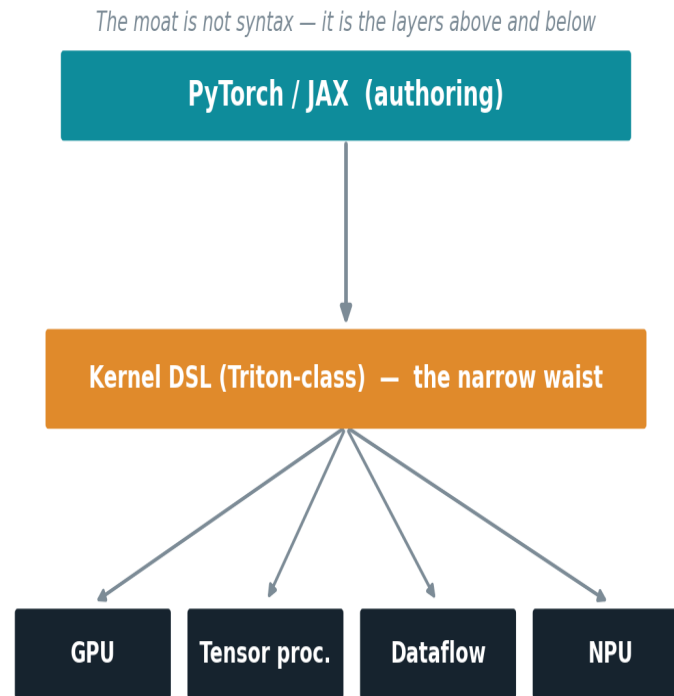
Same families as the datacenter — different binding constraints.

06

Substrate & economics

Software capital, packaging, and cost

The moat is not the kernel language



⇒ Portability efforts aimed at the kernel layer solve the cheapest part; the moat sits above and below the waist.

Integration is the frontier



HBM

Sets every bandwidth bound; three-vendor supply; rivals the logic die in cost share.



Packaging

Reticle limit drives 2.5D, chiplets, hybrid bonding; capacity itself gates supply.

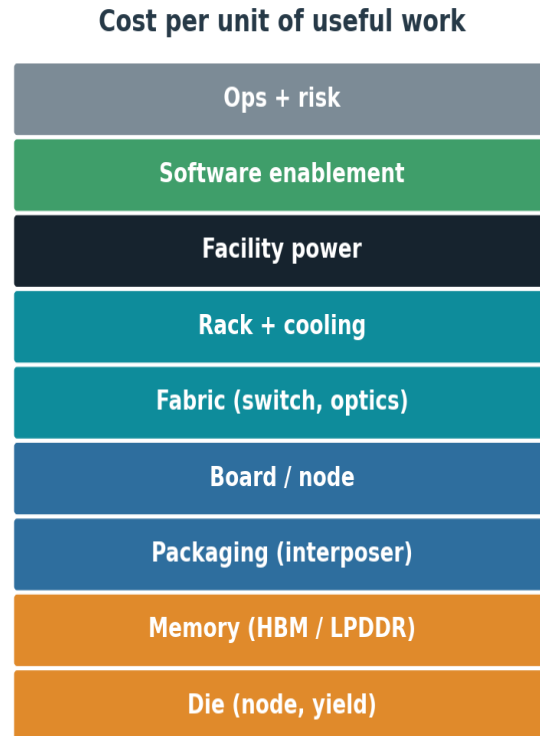


Fabric & optics

Scale-up vs Ethernet; co-packaged optics attacks network power, not latency.

⇒ The competitive frontier moved from logic to integration — the package, the rack, the facility.

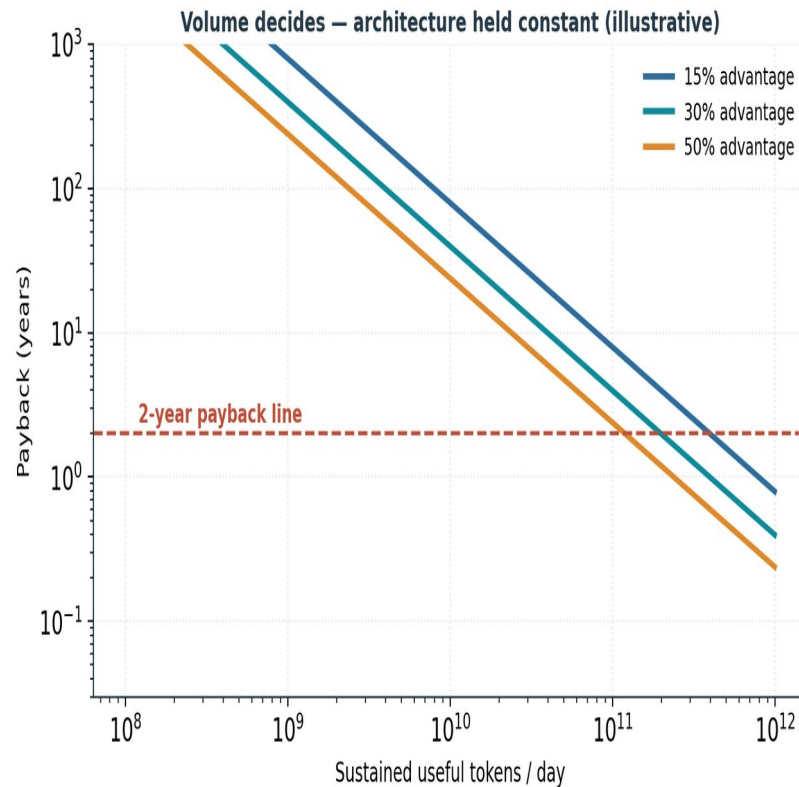
The full cost stack of useful work



- Every family redistributes weight.
- None shrinks the stack uniformly.
- Power now gates deployments.

⇒ The necessity test, in cost terms: does a redistribution lower the total for this workload at this volume?

Volume decides — architecture held constant



Fixed cost $\approx$ software enablement.

Savings scale with volume.

Flexibility has option value.

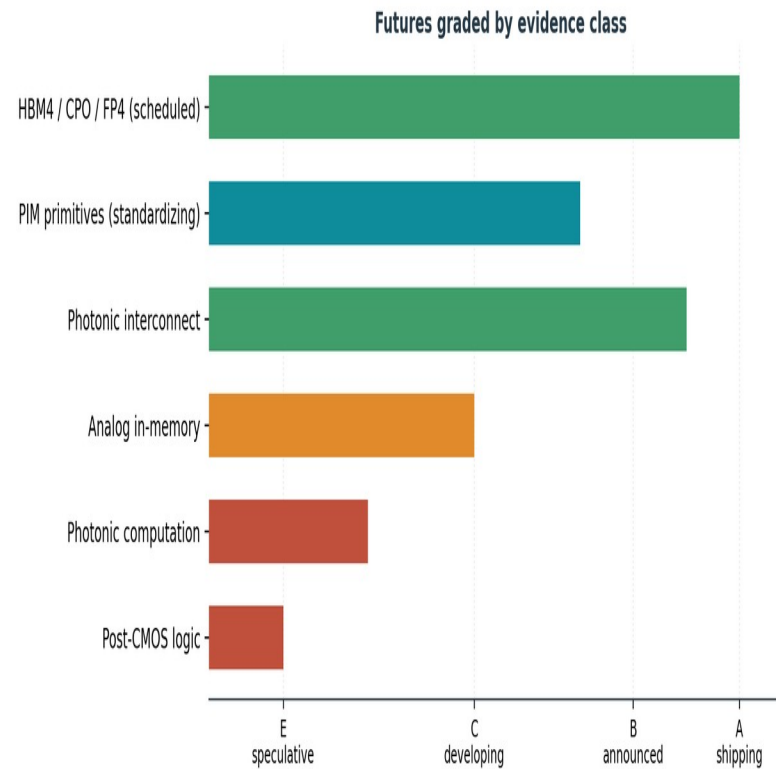
⇒ Below the volume threshold no efficiency percentage rescues the case; above it, small percentages justify large programs.

07

Futures & conclusions

Graded by evidence; what holds everywhere

Graded by evidence class



⇒ New memories arrive before new logic; the nearest contract changes enter inside incumbent platforms, via packaging.

What could move the bounds first



Test-time compute

Reasoning longer, not larger — deepens the decode bound, strengthens inference silicon.



Persistent-state agents

Promote state residency and freshness to first-class bounds — a possible new family.



Default sparsity

MoE and retrieval push routing and gather up the priority list.

⇒ If the signatures move, the family taxonomy should be re-derived — the method predicts its own revision.

What holds across the whole picture

A

Software converges on physics

Both transitions are software succeeding until it exposes a hard, measurable bound.

B

Necessity is decidable

A four-condition test returns specialize / market-no / stay-on-GPU from measurable inputs.

C

Condition 4 dominates

Software capital, volume, and distribution decide outcomes more than silicon merit.

D

Families, not a successor

Each changes a contract the GPU cannot adopt without ceasing to be a GPU.

E

Bounds moved to memory & power

Capacity, bandwidth, fabric, and facility power separate contenders — not FLOPS.

F

One continuum

The same taxonomy reads microwatt silicon and megawatt pods.

How to use the framework



Software architects

Design against signatures; respect batch-one; measure at the useful-result boundary; the test, not the catalog.



Platform architects

Evaluate at pod and facility; prefer consumption to ownership until volume proves otherwise; protect software seams.



Hardware selection

Signatures → software ceiling → locate the bound in the partition → weight the stack → run the test.

THE ARGUMENT IN ONE SENTENCE

Software optimization succeeds until it exposes a bound; the bound is absorbed by the GPU platform, escaped by changing a contract, or lived with — and the deployment class decides which.

Everything else — signatures, contracts, families, economics — is the machinery that makes that choice decidable.